

Vinay Kumar

Systems Engineer Candidate | Autonomous Systems, Telemetry Ingestion, Modern C++, Linux Internals, Python & Distributed Infra

New York, NY | vinaykumarmd02@gmail.com • vk2771@nyu.edu • LinkedIn • GitHub

Target Role: 2027 Systems Engineer | **Degree:** M.S. CS at NYU Tandon (GPA: 3.67 / 4.0) | **Graduation:** May 2027
EDUCATION

New York University (NYU), Tandon School of Engineering

Brooklyn, NY

Master of Science in Computer Science | Cumulative GPA: 3.67 / 4.0

Expected Graduation: May 2027

– **Core Systems & Algorithmic Coursework:** Distributed Systems, Operating Systems, Advanced Algorithms, Database Systems, Machine Learning for Signal Processing, Deep Learning Systems.

B.N.M. Institute of Technology

Bangalore, India

Bachelor of Engineering in Artificial Intelligence & Machine Learning | CGPA: 8.69 / 10

Graduated Apr 2025

EXPERIENCE

Healium Intelliscan Corporation

Bengaluru, India

Software Systems & Telemetry Ingestion Intern

Mar 2025 – Jun 2025

- Engineered high-throughput telemetry ingestion and signal processing pipelines in Modern C++ and Python (FastAPI) to stream and evaluate 80k+ real-time sensor frames with deterministic sub-millisecond budgets.
- Architected non-blocking circular ring buffers and asynchronous worker pools to decouple raw network socket listeners from downstream analysis models, preventing buffer overruns and cutting pipeline latency by 79%.
- Profiled system latency, CPU cache utilization, and memory allocations on Linux hosts using GDB, Valgrind, and ‘perf’, eliminating thread contention and memory leaks under continuous operation.

KreditBee

Bengaluru, India

Software Engineering & Distributed Systems Intern

Sep 2024 – Dec 2024

- Constructed high-throughput transactional verification microservices in Python and PostgreSQL to process high-concurrency event streams under strict latency SLAs, cutting decision latency by 88%.
- Engineered statistical anomaly detection routines over streaming transaction payloads to reason about untrusted input and structural inconsistencies, suppressing false validation triggers by 76%.
- Optimized SQL query plans, indexing strategies, and database connection pools across high-volume operational tables, preventing deadlocks and maintaining ACID compliance under peak write bursts.

NYU Tandon School of Engineering

Brooklyn, NY

Graduate Academic Operations Lead & Systems Specialist

Jan 2026 – Present

- Administer technical lab infrastructure, imaging, and network integrity for CS cohorts of 150+ engineers, debugging OS-level crashes, runtime faults, and hardware access errors under strict operational SLAs.
- Automated cross-departmental reporting workflows by integrating institutional REST APIs via Python and Google Apps Script, eliminating manual tracking overhead for multidisciplinary leads.

PROJECTS

Distributed Strategy Simulation & Real-Time Ingestion Engine

Modern C++ (C++17/20), Python, MATLAB/Simulink Concepts, ZeroMQ, Redis Pub/Sub, Multi-Threading, Linux

- Architected an event-driven distributed processing engine to stream, normalize, and match high-frequency telemetry transaction logs across parallel compute nodes under simulated operational stress.
- Designed an asynchronous worker pool using ZeroMQ (PUSH/PULL) and Redis Pub/Sub, separating ingestion sockets from computationally heavy simulation, state tracking, and reconciliation logic.
- Implemented thread-safe state reconciliation using POSIX mutexes and atomic operations, preventing race conditions and maintaining deterministic memory alignment under concurrent execution.
- Profiled system latency and cache performance on Linux using ‘perf’ and Valgrind, optimizing tight loops to reduce instruction cache misses and branch mispredictions.

Autonomous System Diagnostics & Telemetry Remediation Agent

Python, LangGraph, Linux Internals, Redis, FastAPI, Docker, Microservices

- Architected an autonomous systems debugging agent utilizing ReAct and Plan-Execute paradigms to parse live distributed system logs, TCP connection metrics, and container health states.
- Engineered an extensible tool-execution engine (REST APIs, dynamic SQL queries, system log parsers) with strict input validation to isolate root causes and synthesize actionable incident post-mortems.
- Constructed a semantic RAG pipeline (FAISS, semantic chunking, cross-encoder reranking) that suppressed context hallucinations by 45% and achieved 75% diagnostic precision across complex incident traces.
- Deployed an asynchronous FastAPI backend paired with Redis caching, maintaining sub-50ms query evaluation under concurrent multi-agent debugging sessions.

PromptDoctor: Developer Evaluation Platform & Secure LLM Governance Workbench

Python, Ruby on Rails, PostgreSQL, Redis, Sidekiq, AI SDLC Toolchain, RBAC

- Built an end-to-end evaluation platform to benchmark prompt ambiguity, structural constraint regressions, and token performance trade-offs across concurrent model iterations.
- Architected asynchronous worker queues via Redis and Sidekiq to handle long-running model evaluations in parallel, persisting versioned benchmark runs in normalized PostgreSQL schemas.
- Enforced strict role-based access control (RBAC), sanitized test inputs, and audit-logged execution traces to maintain data confidentiality and governance across multi-tenant test datasets.

SKILLS

Systems & Languages: Modern C++ (C++17/20), Python, Go (Golang), MATLAB Concepts, SQL (PostgreSQL, MySQL), Bash
OS, Telemetry & Linux: Linux Internals, POSIX Sockets, Concurrency (Atoms, Mutexes), Ring Buffers, Sensor Fusion Concepts
Distributed Systems: ZeroMQ, Redis (Pub/Sub & Cache), Microservices, Event-Driven Architectures, Docker, Pydantic
DevOps & Tools: GDB, Valgrind, ‘perf’, Git/GitHub, CI/CD, AWS (EC2, S3), Claude Code, Model Context Protocol (MCP)

ACHIEVEMENTS & CERTIFICATIONS

- **Patent Contributor:** Co-inventor on an IoT sensor tracking patent for automated real-time telemetry capture and threshold control.